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THE DIPOLE MOMENT OF ^{7}LiH AND ^{7}LiD IN THE EXCITED $A\,^1\Sigma^+$ STATE: A TEST OF THE BORN–OPPENHEIMER APPROXIMATION ☆

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Continuing our Stark quantum beat experiments on ^{7}LiH at electric field strengths of 100 kV/cm and above, we were able to observe quantum beat signals for the vibrational level $v' = 5$ of the $A\,^1\Sigma^+$ state. The evaluation of these signals yields a field reduced splitting $\Delta W/E^2 = 1.157(70)$ kHz/(kV/cm)^2 which would correspond to an "effective" value of $|\mu_{\text{eff}}(v' = 5)| = 0.051(5)$ D for the dipole moment, if the ordinarily used formula were applicable. A serious check of all available data, however, shows that neighbouring vibrational levels of the $A\,^1\Sigma^+$ state and also the sum over the levels of the electronic ground state $X\,^1\Sigma^+$, though counteracting, have the main effect on the residual splitting thus preventing a straightforward evaluation for the dipole moment. Taking account of these polarizability effects and including the possibility for a slight shift of the theoretical dipole moment function a value of $|\mu(v' = 5)| = 0.089(14)$ D is the most probable one. In the case of ^{7}LiD according to the usual formula an "effective" value of $|\mu_{\text{eff}}(v' = 6)| = 0.425(20)$ D was deduced from a beat signal obtained at an electric field strength of 24 kV/cm. Here the influence of neighbouring vibrational levels and electronic states is only of minor importance leading to a corrected value of $|\mu(v' = 6)| = 0.43(2)$ D for the dipole moment. This value differs by 13% from the theoretical one calculated on the basis of the dipole moment function of ^{7}LiH and the adiabatically corrected potential energy curve of ^{7}LiD, thus indicating non-adiabatic effects.

1. Introduction

Due to some remarkable features alkali hydrides have attracted attention of many investigations, both theoretical [1–3] as well as experimental [4–6] ones, ever since. One of these features originating from avoided crossings between zero-order covalent and ionic potential energy curves, is the strongly deformed potential energy curve of the first electronically excited state $A\,^1\Sigma^+$. An outstanding property of alkali hydrides in this state is the pronounced dependence of the electric dipole moment on the vibrational quantum number which is inferred from the strongly R-dependent dipole moment function. This behaviour was predicted theoretically [1] and recently also confirmed experimentally for LiH [7,8] and NaH [9]. In the case of LiH and LiD a further peculiarity may be seen in the fact that due to a drastic change in the composition of the electronic wavefunction, the dipole moment changes sign as it runs through the different vibrational levels of the $A\,^1\Sigma^+$ state. The zero crossover point where contributions of the negative and the positive branch of the electric dipole moment function cancel on vibrational averaging is nearly exactly reached for ^{7}LiH in the vibrational level $v' = 5$, while for ^{7}LiD the level $v' = 7$ is the one closest to this point. So in principle, measuring the electric dipole moments of the levels $v' = 5$ and $v' = 7$ for ^{7}LiH and ^{7}LiD, respectively, as accurately as possible would offer the chance of revealing even small deviations from theoretical predictions of ab initio MC SCF calculations [1–3], if there were not the polarizabilities that cover these effects. The ab initio calculations are based on the Born–Oppenheimer approximation which claims identical potential energy curves as well as electric dipole moment functions for LiH and LiD. It has, however, been demonstrated that the Born–Oppenheimer approximation is not sufficient for an exact descrip-

☆ This work is dedicated to Professor G. Ludwig on the occasion of his retirement from Philipps-Universität, Marburg, FRG.

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tion of these molecules, at least as far as energy eigenvalues are concerned: Using the most reliable spectroscopic data, Vidal and Stwalley [10] have been able to give remarkably improved fits to all line positions of the four stable isotopic species ^{6,7}LiH and ^{6,7}LiD, when introducing adiabatic corrections to the common Born–Oppenheimer potential energy curve of the isotopic combinations. This established the need for the adiabatic approximation when describing these molecules on a general footing.

Being a first-order perturbation of the electronic energy by the motion of the nuclear frame, the adiabatic corrections alter the common Born–Oppenheimer potential energy curve differently for the isotopic species and thus the vibrational wavefunctions by more than due to their dependence on the reduced mass, but leave the fixed-center electronic wavefunctions unaffected. The dipole moment function is therefore the same in Born–Oppenheimer and adiabatic approximation. The observable dipole moment is given by the expectation value of the actual dipole moment function with the wavefunction of the vibrational level involved. In this way, experimentally determined dipole moments of LiH and LiD e.g. can provide landmarks for the comparison with theoretically deduced dipole moments thereby probing the validity of even the adiabatic approximation.

Any experimental determination of electric dipole moments, however, generally starts with measuring electric-field-induced splittings of energy levels. For molecular $^1\Sigma$ states the perturbation of lowest order causing such splitting, is the second-order Stark effect which inherently involves a weighted sum over non-diagonal dipole matrix elements. Usually only two terms are strongly dominating in the sum and have to be considered. The matrix elements in these terms describe the interaction with the adjacent rotational levels. The integration over the eulerian angles is easily done and, if the dependence of the vibrational wavefunctions on different rotational quantum numbers is negligible in the integrals, the remainder of the vibrationally diagonal matrix elements can be approximated by the electric dipole moment.

In the case of the $A\,^1\Sigma^+, v' = 5, J' = 1$ level of ^{7}LiH, however, as will be discussed below and in ref. [11], the purely rotational terms not only fail to be dominating, but also show a non-negligible dependence of the rotationally reduced matrix elements on rotation. Furthermore, vibrationally and electronically non-diagonal dipole matrix elements prove to be very important thus covering finer effects and making the analysis involved. As will be shown in ref. [11], all these non-diagonal contributions can be interpreted in terms of polarizabilities.

2. Theory

For a $^1\Sigma$ state the energy shift W_α of a molecular level induced by an electric field may be treated by second-order perturbation theory as long as it is small compared to the level spacings $E_\alpha - E_\beta$:

$$W_\alpha = \sum_{\beta \neq \alpha} |H_{\alpha\beta}|^2 / (E_\alpha - E_\beta),$$

where α and β are appropriate sets of molecular quantum numbers. The summation runs over all molecular levels and states β with energy E_β which are connected to the perturbed level α with energy E_α by the perturbing hamiltonian

$$H' = -\boldsymbol{\mu} \cdot \boldsymbol{E},$$

where

$$\boldsymbol{\mu} = e\left[\sum_k Z_k R_k - \sum_i r_i\right]$$

is the dipole moment operator given by the nuclear charges Z_k, nuclear coordinates R_k and electronic coordinates r_i, respectively, interacting with the external electric field E.

For the $A\,^1\Sigma^+$ state of ^{7}LiH, as will be explicitly derived in ref. [11], the energy shift of the level described by the quantum state $|A\,^1\Sigma^+, v, J, M\rangle$ is given by summing over all contributions of interacting rovibronic levels of all $^1\Sigma$ and $^1\Pi$ states. Due to the limited knowledge of the internal structure of this molecule account can only be taken of the few known electronic states:

$$W^{(2)}_{A^1\Sigma, v(J), J, M} = E^2 \sum_{J'} (2J'+1)(2J+1) \begin{pmatrix} J' & J & 1 \\ M & -M & 0 \end{pmatrix}^2 \times \left\{ \begin{pmatrix} J' & J & 1 \\ 0 & 0 & 0 \end{pmatrix}^2 [(\mathrm{I}) + (\mathrm{II}) + (\mathrm{III})] + \begin{pmatrix} J' & J & 1 \\ -1 & 0 & 1 \end{pmatrix}^2 (\mathrm{IV}) + \ldots \right\}, \quad (1)$$

where

$$(\mathrm{I}) = \frac{|\langle A\,^1\Sigma, v(J)|\mu(R)|A\,^1\Sigma, v(J')\rangle|^2}{G_{v+1/2}(J) - G_{v+1/2}(J')},$$

$$(\mathrm{II}) = \sum_{v' \neq v} \frac{|\langle A\,^1\Sigma, v(J)|\mu(R)|A\,^1\Sigma, v'(J')\rangle|^2}{G_{v+1/2}(J) - G_{v'+1/2}(J')},$$

$$(\mathrm{III}) = \sum_{v''} \frac{|\langle A\,^1\Sigma, v(J)|D_{AX}(R)|X\,^1\Sigma, v''(J')\rangle|^2}{W^{(0)}_{A\,^1\Sigma,v,J} - W^{(0)}_{X\,^1\Sigma,v'',J'}},$$

$$(\mathrm{IV}) = \sum_{v'} \frac{|\langle A\,^1\Sigma, v(J)|D_{AB}(R)|B\,^1\Pi, v'(J')\rangle|^2}{W^{(0)}_{A\,^1\Sigma,v,J} - W^{(0)}_{B\,^1\Pi,v',J'}},$$

and ... in (I) stands for contributions of unknown electronic states to the polarizabilities of the A state.

The integration over the electronic coordinates has already been carried out, leading to the dipole moment function

$$\mu(R) = \langle \psi_A(r, R)|\mu_0^M|\psi_A(r, R)\rangle$$

and to the transition dipole moment functions

$$D_{AB}(R) = \langle \psi_A(r, R)|\mu_{-1}^M|\psi_B(r, R)\rangle,$$

$$D_{AX}(R) = \langle \psi_A(r, R)|\mu_0^M|\psi_X(r, R)\rangle.$$

μ_q^M is a spherical vector component of the dipole moment operator in the molecule-fixed system of reference.

Due to the centrifugal barrier the potential energy curves are slightly different for different rotational quantum numbers J giving rise to non-identical vibrational wavefunctions for the same vibrational level as characterized by $v(J)$ and $v(J')$ and to different vibrational energies $G_v(J)$ and $G_v(J')$. The difference of these energies is normally approximated by the difference of the rotational energies $B_v[J(J+1) - J'(J'+1)]$, B_v being the rotational constant of the vibrational level.

In this way, the shift of a molecular energy level belonging to a $^1\Sigma$ state which is induced by an electric field consists of four contributions caused by:

(I) interaction with adjacent rotational levels;

(II) interaction with rovibrational levels of the same electronic $^1\Sigma$ state;

(III) interaction with rovibrational levels of other electronic $^1\Sigma$ states;

(IV) interaction with rovibrational levels of electronic $^1\Pi$ states.

Usually, the vibrational averaging of the dipole moment function is far from being close to zero, and the contribution of the first part (I) is large compared to the other ones. Then, if these small polarizability contributions (II)–(IV) and the J dependence of the matrix element can be neglected, the well-known expression for the electric-field-induced splitting of a rotational level belonging to a $^1\Sigma$ state is obtained:

$$\Delta W^{(2)n\,^1\Sigma,v}_{n\,^1\Sigma,v(J),J,M',M} \approx \frac{\mu^2_{v(J)}F^2}{2B_v^{n\,^1\Sigma}J(J+1)} \frac{J(J+1) - 3[M^2 - (M')^2]}{(2J-1)(2J+3)}. \quad (2)$$

In the case of the $A\,^1\Sigma^+$, $v' = 5$, $J' = 1$ of ^{7}LiH the almost complete cancellation of positive and negative contributions in the vibrational averaging of the dipole moment function makes this integration very sensitive to small variations in any of the functions constituting the matrix element. Ignoring the influence of rotation on the vibrational wavefunction even at the low rotational quantum numbers involved, $J = 0$, 1 and 2, would bring about an error of 6.5% in the fractional splitting contributed by part (I) above. Therefore it does not seem reasonable to make use of the approximation leading to formula (2) above, especially since parts (II) and (III) in formula (I) prove to be the most important, but counteracting contributions.

3. Experimental

The electric-field-induced splittings in the energy levels $A\,^1\Sigma^+$, $J' = 1$, $v' = 5$ and $v' = 6$ of ^{7}LiH and ^{7}LiD, respectively, have been measured by applying the technique of laser-induced Stark quantum beats using the R(0) line for excitation. This method allows exact determinations of such level splittings in electronically excited molecular states from which electric dipole moments usually can be readily deduced. The details of the experimental arrangement are the same as described in ref. [8]. So we would like to call attention to some special features of the experiments reported here.

As is obvious from the excitation spectrum of the band head of the (5,0) band in the A–X system of ^{7}LiH, depicted on top of fig. 1, the bandwidth of the

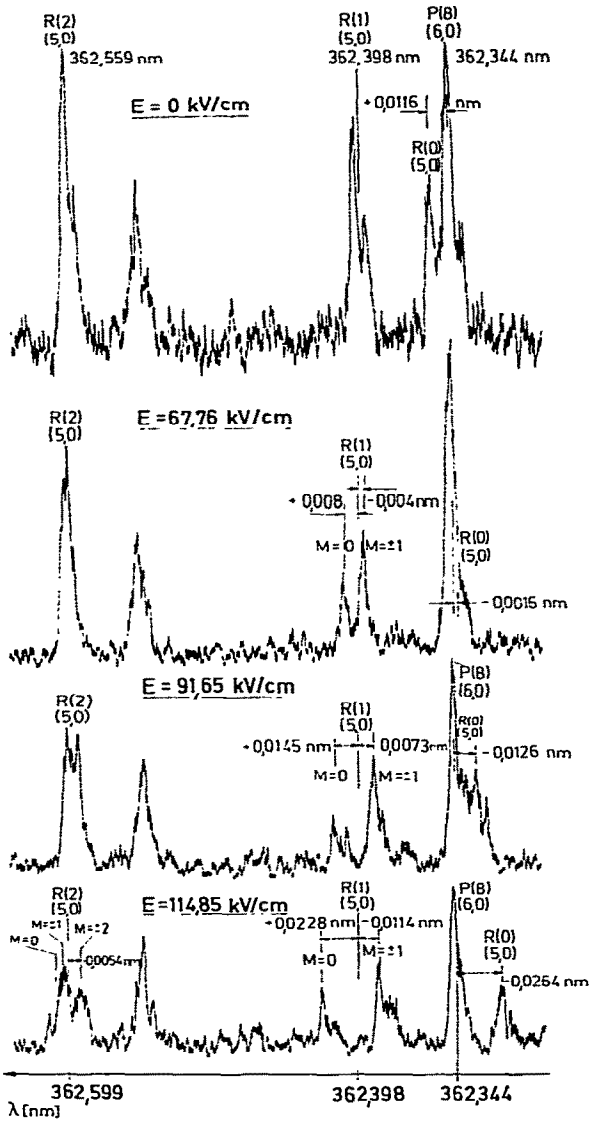


Fig. 1. Excitation spectra showing the (5,0) band head of the A–X transition in ^{7}LiH and the electric field shift of the R(0) line at different field strengths. At 92 kV/cm the R(0) line is already shifted to the blue side of the (6,0), P(8) line.

nitrogen-laser pumped dye laser oscillator–amplifier system (run with 5×10^{-3} mol PBD per liter toluene) was well below 0.01 nm, since the lines (5,0), R(0) and (6,0), P(8) being 0.0116 nm apart, are clearly resolved. This spectrum, however, as shown in the other excitation spectra of fig. 1, changes dramatically when the LiH molecules are exposed to electric field strengths of the order of 100 kV/cm. Then, for instance at 115 kV/cm (bottom of fig. 1), due to the large electric dipole moment of 5.88 D [12] in the $X\,^1\Sigma^+$, $v'' = 0$ ground state the energy of the rotational level $J'' = 0$ ($B_{v''=0} = 7.4067\ \mathrm{cm}^{-1}$ [10]) is lowered by as much as $2.9\ \mathrm{cm}^{-1}$, giving rise to a shift of the R(0) line of 0.038 nm towards shorter wavelengths. At this field strength the R(0) line appears well resolved at the blue side of the (6,0), P(8) line, which because of the high J'' value remains practically stationary. Fig. 1 illustrates the shift of the (5,0), R(0) line with respect to the (6,0), P(8) line for three markedly different field strengths. At 63.4 kV/cm both lines coincide theoretically, as do the experimental lines at 68 kV/cm with the spectral resolution achieved. For getting the (5,0), R(0) line well isolated from the (6,0) P(8) line a field strength of at least 80 kV/cm should be applied. The generation of such high field strengths was accomplished by diminishing the separation of the capacitor plates to 0.565(10) cm and applying voltages of the order of 60 kV.

4. Results

Fig. 2 shows a decay signal of the $A\,^1\Sigma^+$, $v' = 5$, $J' = 1$ level of ^{7}LiH distorted by a low-frequency beat which was obtained at a field strength of 99.6 kV/cm during a data aquisition time of 1.5 h. The result of the fit, displayed by the solid line in the signal is ob-

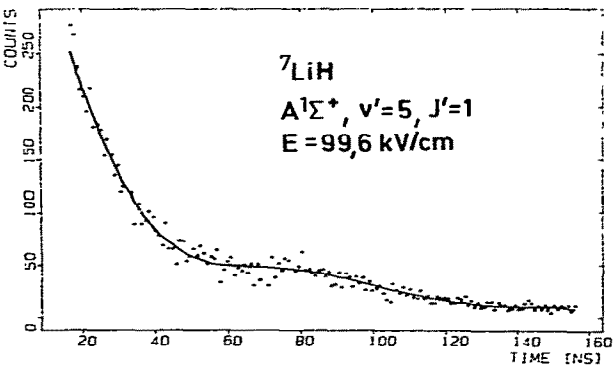


Fig. 2. Quantum beat signal of the $A\,^1\Sigma^+$, $v' = 5$, $J' = 1$ level of ^{7}LiH obtained at a field strength of 99.6 kV/cm. The solid line represents a computer optimized fit yielding a beat frequency of 11.4 MHz.

tained for a beat frequency of 11.4(8) MHz. From another measurement at 115.8 kV/cm a beat frequency of 15.6(5) MHz was obtained. Because of the comparatively slow oscillations the fit errors amount to 7.0% and 3.2%, respectively. For keeping these errors as small as possible the lifetime was not subjected to the fitting procedure in this case, but taken fixed, as determined from an independent decay experiment to be $\tau = 35.2(1.5)$ ns [13]. Although admixtures to the wavefunction occur already in first order, proceeding in this way is justified by the negligible amount of these admixtures even at the very high field strengths applied: At 100 kV/cm the transition probability is changed by less than 2×10^{-3}. This is much less than the uncertainties involved in the experimental determination of the lifetime. Further sources of uncertainties were:

3% for the determination of the electric field strengths;

1.0% for the absolute time calibration.

So the relative overall error in the measurement of the field reduced splitting $\Delta W/E^2 = 1.157(70)$ kHz/(kV/cm)^2 is 6% for the $v' = 5$ level of ^{7}LiH.

Within the limits of the experimental errors the ratio of the beat frequencies ν(115.8 kV/cm)/ν(99.6 kV/cm) = 1.37 corresponds very well to the value of 1.35 which would be expected from an E^2 dependence. This rules out higher-order effects that are larger than the sum of the experimental uncertainties.

For taking into account the influence of neighbouring rovibrational levels of the $A\,^1\Sigma^+$ state and of the other known electronic states $X\,^1\Sigma^+$ and $B\,^1\Pi$ on the electric field splitting [see eq. (1)] the best data available at present were used:

the theoretical dipole moment function of ref. [3];

an adiabatic potential energy curve for the $A\,^1\Sigma^+$ state of ^{7}LiH obtained by means of the IPA method [10];

electronic transition probabilities of refs. [14,15].

Fig. 3 demonstrates how the predicted electric field reduced level splitting is composed from the contributions of the rotational, vibrational and electronic levels or states when using these data. It is clearly established that compared to the normally dominating part (I) of formula (1) the other rovibrational levels of the $A\,^1\Sigma^+$ state (part II) as well as especially those of the electronic ground state $X\,^1\Sigma^+$ (part III) yield large contributions to the Stark splitting, but cancel

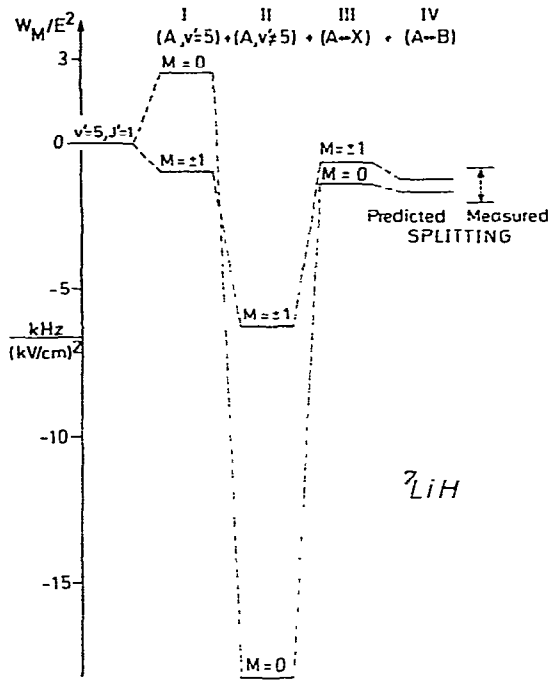


Fig. 3. Theoretical contributions to the splitting of the $A\,^1\Sigma^+$, $v' = 5, J' = 1$ level of ^{7}LiH. The contributions arising from the rovibrational (II) and rovibronic parts (III, IV) of the interaction are essentially determining the electric field splitting.

each other almost completely. The resulting theoretical field reduced splitting, however, which is to be compared with the measured one, turns out to be too small by a factor of 2.64. Reasons for this discrepancy will be discussed roughly in section 5 and in more detail in ref. [11].

In the case of ^{7}LiD no such high field strengths were necessary for obtaining beat signals from the $A\,^1\Sigma^+$, $v' = 6, J' = 1$ level. Fig. 4 depicts such a signal the data of which had been accumulated during 1 h at a field strength of 24 kV/cm. This field strength was easily attained with a plate separation of 1.00(1) cm. The signal exhibits a well-developed modulation with a frequency of 80.6 MHz corresponding to an "effective" electric dipole moment of 0.425(20) D [eq. (2)]. Fig. 5 depicts the theoretical contributions to the electric field reduced splitting of the $v' = 6, J' = 1$ level in the $A\,^1\Sigma^+$ state of ^{7}LiD, as calculated according to expression (1). The main contribution to the level splitting arises from the dipole moment (part I) of this

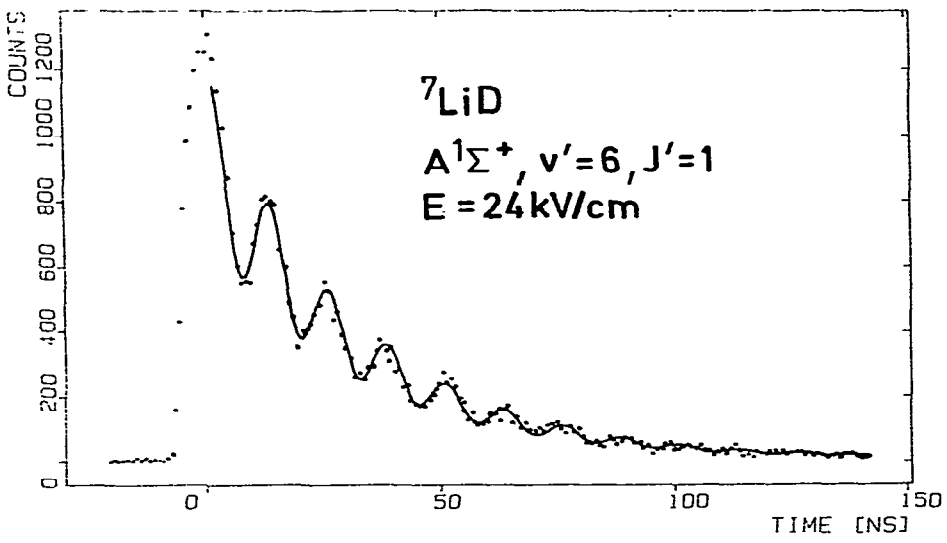


Fig. 4. Quantum beat signal of the $A\,^1\Sigma^+$, $v' = 6, J' = 1$ level of ^{7}LiD obtained at a field strength of 24 kV/cm. The nicely developed beat structure keeps the fit error small and yields a beat frequency of 80.6 (8) MHz.

level, but the influence of the other levels $v' \neq 6$ of the $A\,^1\Sigma^+$ state and of the levels of the $X\,^1\Sigma^+$ ground state leads to a slightly diminished predicted energy

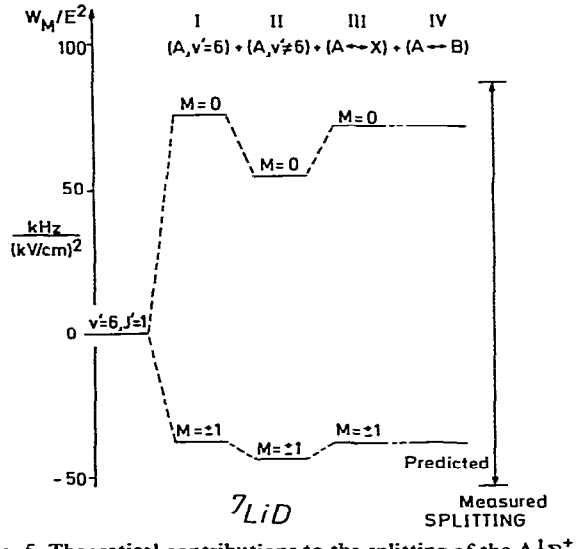


Fig 5. Theoretical contributions to the splitting of the $A\,^1\Sigma^+$, $v' = 6, J' = 1$ level of ^{7}LiD. Due to the larger dipole moment in this level the first part (I) of formula (1) as approximated by formula (2) which reflects purely rotational interaction, is dominating. The other contributions lead to a small correction of the dipole moment.

splitting. It should be mentioned at this point that since no theoretical data exist for the A–X and A–B transition dipole moments or ^{7}LiD, we approximated the influence of the ground state and of the $B\,^1\Pi$ state of ^{7}LiD by the same sums that were used for ^{7}LiH. This approximation is justified by the following facts:

(1) the Born–Oppenheimer approximation is sufficiently good for this purpose;

(2) the lifetimes for the A-state levels $v' = 5$ of ^{7}LiH and $v' = 6$ of ^{7}LiD turn out to be very similar [13];

(3) the influence of these contributions on the overall splitting is only of minor importance.

In this case as well the theoretical splitting is smaller than the experimental one, this time by a factor of 1.28. Attributing this deficit totally to the dipole moment function of the A state we arrive at a corrected value of $|\mu(v' = 6)| = 0.43(2)$ D for the electric dipole moment of ^{7}LiD.

The experimental and theoretical values of dipole moments in the $A\,^1\Sigma^+$ states of ^{7}LiH and ^{7}LiD are summarized in table 1. For ^{7}LiH the experimental data agree very well with theoretical results computed by means of the dipole moment function of ref. [3] and the IPA potential energy curve [10] which to our conviction are the best data available at present.

The theoretical dipole moment of ^{7}LiD was calculated by means of the dipole moment function of ^{7}LiH

Table 1
Comparison of theoretical and experimental values for the dipole moments in some vibrational levels of the $A\,^1\Sigma^+$ state of ^{7}LiH and ^{7}LiD. Theoretical dipole moments have been obtained by averaging the dipole moment function of ref. [3] with adiabatically corrected vibrational wavefunctions

v'	G_v (cm^{-1})	$\lvert\mu$(theor.)$\rvert$ (D)	$\lvert\mu$(exp)$\rvert$ (D)	
			ref. [7]	this work and ref. [8]
^{7}LiH 1	412.24	2.03	1.88 (19)	
2	725.20	1.52	1.48 (18)	1.46 (7)
4	1413.73	0.55		0.56 (4)
5	1779.56	0.085	0.29 (14)	0.089 (14)
^{7}LiD 6	1549.45	0.38		0.43 (2)

and the IPA potential energy curve of the $A\,^1\Sigma^+$ state of ^{7}LiD. The experimental value differs significantly from the theoretical result indicating that the dipole moment functions of LiH and LiD are slightly different.

5. Discussion

The ab initio MC SCF calculations of Partridge et al. [3] have shown the electric dipole moment of the $A\,^1\Sigma^+$, $v' = 5$ level of ^{7}LiH to be very sensitive to minor modifications of either the potential energy curve or the electric dipole moment function with respect to one another. It should be kept in mind, however, that this sensitivity is due to an almost total cancellation of large negative and positive contributions to the vibrational average of the dipole moment function, as pointed out above. In this sense, a very small shift of the electric dipole moment function either with respect to the potential energy curve or in magnitude will result in a relatively large variation of the expectation value for the dipole moment of the $v' = 5$ level.

The agreement between the results of ab initio MC SCF calculations and the experimental data depends very much on the choice of the basis sets, as has been pointed out in ref. [3]. The best basis sets, however, cannot account for all details when used in an inadequate frame: Using the inverted perturbation approach (IPA) first proposed by Kosman and Hinze [16] which is a purely quantum-mechanical variational method, Vidal and Stwalley [10] have calculated Dunham-type coefficients for the molecular constants and the potential energy curves from all experimental data available for the $X\,^1\Sigma^+$ and $A\,^1\Sigma^+$ states of all naturally occurring isotopic combinations ^{6,7}LiH and ^{6,7}LiD. Their results clearly demonstrate that the Born–Oppenheimer approximation is not capable of accounting for finer details in the spacings of the energy eigenvalues and that a drastic amendment in the fits of the line positions can be achieved when incorporating adiabatic corrections for both states. Being more important for the $A\,^1\Sigma^+$ state these corrections turn out to be significantly different when hydrogen is substituted by deuterium, while on the other hand they are rather insensitive to the substitution of the lithium isotopes. On replacing hydrogen by deuterium significant changes occur in the shape of the X- and A-potential energy curves as well as for the electronic term difference.

On the basis of the theoretical dipole moment function of ref. [3] and the most accurate experimental potential energy curves, as obtained by the IPA method [10], theoretical values for the dipole moments of ^{7}LiH and ^{7}LiD have been calculated for those vibrational levels of the $A\,^1\Sigma^+$ state for which experimental data are available. The results are listed in table 1 and displayed in fig. 6. The very good agreement for ^{7}LiH indicates that for this molecule the dipole moment can be predicted quite accurately by the adiabatically correct vibrational averaging of the theoretical dipole moment function as far as the investigated range of vibrational levels is concerned. As mentioned in section 4 the experimentally determined splitting of the $A\,^1\Sigma^+$, $v' = 5, J' = 1$ level turns out to be 2.64 times larger than the theoretically predicted one. Possible reasons for this discrepance will be discussed in detail in ref. [11]. This discussion can be summarized as follows: One of the reasons may be, of course, that only the contributions of the known states to the polarizabilities of the investigated level have been taken into account. If these contributions have the correct magnitude, then, in fact, the missing portion has to be attributed to the influence of unknown electronic states Since, on the other hand, reducing the influence of the strongly contributing $X\,^1\Sigma^+$ state by only 6.4% is sufficient for totally explaining the measured splitting, and since our lifetime measurements [13] indi-

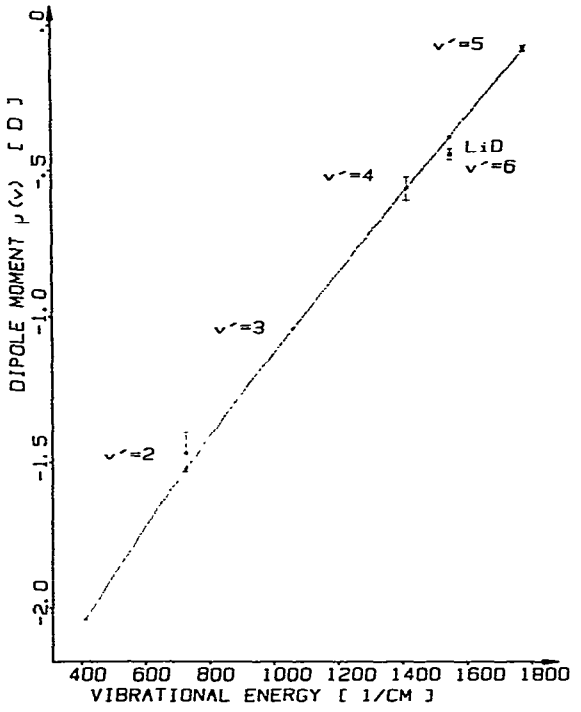


Fig. 6. Comparison of theoretical with experimental values of dipole moments. There is quite a good agreement for ^{7}LiH whereas the experimental value for ^{7}LiD differs significantly from the theoretical one.

cate that the theoretical transition probability is too large by 10%, the need for an alteration of the X-contribution cannot be excluded. All these alterations do not affect the theoretical assumptions for the A state. As far as the contributions from the vibrational levels of the A state are concerned, to our conviction the accuracy of the available data makes these contributions the least probable ones for a possibly necessary amendment. Nevertheless, even if such an amendment were necessary, the worst that could happen to the data for the A state, is that they had to account for the missing splitting alone. The easiest way of making them do so is to allow a shift in the magnitude of the dipole moment function: $\mu_{corr.}(R) = \mu_{th.}(R) + \delta$. With the adiabatic vibrational wavefunctions applied the result of a quadratic equation gives two possible shifts $\delta_1 = +0.0093$ D and $\delta_2 = +0.176$ D, which would alter the theoretical dipole moment $\mu(v' = 5) = -0.0849$ D by the respective amount. The second shift δ_2 of nearly +0.2 D, however, is ruled out by the accuracy of the dipole moments for the other vibrational levels (see table 1). In case the level order were inverted for the measured splitting a different pair of shifts would be obtained, $\bar{\delta}_1 = -0.0179$ D and $\bar{\delta}_2 = +0.2034$ D. Here the large second shift can be skipped because of the same argument as above. Although principally the sign of a splitting can be determined by adapting the polarization of the exciting and observed light for a sine-beat, the sign of the dipole moment cannot be determined in second-order Stark effect. Without questioning the sign of the theoretical dipole moment the possible shifts hence give upper and lower limits only for the absolute value of the dipole moment: $0.0756\ \text{D} \leqslant |\mu(v' = 5)| \leqslant 0.1028$ D. The value quoted in table 1 is the mean of the limits given above while the error accounts for all possibilitities.

The measurement of the dependence of the electric field splitting on the rotational quantum number J' should give more information on the polarizabilities. But even higher field strengths would be necessary in order to obtain signals with a modulation structure which allows a sufficiently accurate evaluation.

The theoretical value of the dipole moment for the $A\,^1\Sigma^+$, $v' = 6$, $J' = 1$ level of ^{7}LiD, as mentioned above, was obtained by using the theoretical dipole moment function of ^{7}LiH and the IPA potential energy curve of ^{7}LiD. Therefore, this theoretical dipole moment incorporates an adiabatic correction which takes account of the change of the potential energy curve when hydrogen is substituted by deuterium. The experimental value, however, differs significantly from the theoretical one. This deviation cannot be explained in the frame of the adiabatic approximation which still claims identical dipole moment functions for all the isotopic combinations. Although the spectra of the lithium hydrides and deuterides can be explained to high accuracy by means of the adiabatic approximation [10] our measurements of dipole moments in the $A\,^1\Sigma^+$ state of ^{7}LiH and ^{7}LiD suggest that non-adiabatic effects have to be considered when electronic properties of these molecules are described.

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References

[1] K.K. Docken and J. Hinze, J. Chem. Phys. 57 (1972) 4928, 4936.
[2] H. Partridge and S.R. Langhoff, J. Chem. Phys. 74 (1981) 2361.
[3] H. Partridge, S.R. Langhoff, W.C. Stwalley and W.T. Zemke, J. Chem. Phys. 75 (1981) 2299.
[4] F.H. Crawford and T. Jorgensen, Phys. Rev. 47 (1935) 932.
[5] K.C. Li and W.C. Stwalley, J. Mol. Spectry. 69 (1978) 294.
[6] F.B. Orth and W.C. Stwalley, J. Mol. Spectry. 76 (1979) 17.
[7] P.J. Dagdigian, J. Chem. Phys. 73 (1980) 2049.
[8] M. Brieger, A. Hese, A. Renn and A. Sodeik, Chem. Phys. Letters 76 (1980) 465.
[9] M. Brieger, A. Hese, A. Renn and A. Sodeik, Chem. Phys. Letters 78 (1981) 153
[10] C.R. Vidal and W.C. Stwalley, J. Chem. Phys. 77 (1982) 883.
[11] M. Brieger, to be published.
[12] L. Wharton, L.P. Gold and W. Klemperer, J. Chem. Phys. 37 (1962) 2149.
[13] M. Brieger, A. Hese, A. Renn and A. Sodeik, to be published.
[14] W.T. Zemke and W.C. Stwalley, J. Chem. Phys. 68 (1978) 4619.
[15] W.T. Zemke, K.R. Way and W.C. Stwalley, J. Chem. Phys. 69 (1978) 402.
[16] W.M. Kosman and J. Hinze, J. Mol. Spectry. 56 (1975) 93.